Strain comparisons % highly related strains (>=99% ANI) % less related strains (<99% ANI)

Milk kefir

Pseudomonas_E lundensis	100	0
Kluyvera cryocrescens	100	0
Pseudomonas_E sp002966775	95.5	4.5
Leuconostoc pseudomesenteroides_A	93.3	6.7
Leuconostoc pseudomesenteroides	93.3	6.7
Pseudomonas_E fragi	92.9	7.1
Bacillus_A mycoides	91.7	8.3
d__Bacteria;p__Firmicutes;c__Bacilli;o__Paenibacillales;f__Paenibacillaceae;g__Paenibacillus;s__	90.5	9.5
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Enterobacteriaceae;g__Kluyvera;s__	87.5	12.5
Acinetobacter johnsonii	84.6	15.4
Lactococcus lactis_E	80.3	19.7
Escherichia flexneri	80	20
Raoultella terrigena	75	25
Pseudomonas_E proteolytica_A	75	25
Pseudomonas_E brenneri_B	75	25
Bacillus_A wiedmannii	75	25
Serratia liquefaciens	66.7	33.3
Paenibacillus sp001955855	66.7	33.3
Leclercia adecarboxylata	66.7	33.3
Lactococcus lactis	64.4	35.6
Pseudomonas_E putida	61.5	38.5
Lactococcus piscium_C	60.8	39.2
Pseudomonas_E helleri	54.5	45.5
Pantoea agglomerans	54.5	45.5
Citrobacter freundii	53.3	46.7
Acinetobacter albensis	53.3	46.7
Lelliottia amnigena	52.5	47.5
Thermus parvatiensis	48.4	51.6
Clostridium beijerinckii	48.3	51.7
Escherichia coli_D	40	60
Rahnella aquatilis_B	22.2	77.8
Pseudomonas_E fragi_B	21.4	78.6
Leuconostoc mesenteroides	18.1	81.9
Carnobacterium maltaromaticum	16.7	83.3
Acetobacter orientalis	14.3	85.7
Acetobacter fabarum	1.1	98.9
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Pseudomonadales;f__Pseudomonadaceae;g__Pseudomonas_E;s__	0	100
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Enterobacteriaceae;g__Superficieibacter;s__	0	100
d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Acetobacterales;f__Acetobacteraceae;g__Acetobacter;s__	0	100
d__Bacteria;p__Firmicutes_A;c__Clostridia;o__Clostridiales;f__Clostridiaceae;g__Clostridium;s__	0	100
d__Bacteria;p__Firmicutes;c__Bacilli;o__Lactobacillales;f__Lactobacillaceae;g__Lentilactobacillus;s__	0	100
d__Bacteria;p__Firmicutes;c__Bacilli;o__Lactobacillales;f__Lactobacillaceae;g__Lactobacillus;s__	0	100
Streptococcus thermophilus	0	100
Streptococcus parauberis	0	100
Pseudomonas_E sp003852405	0	100
Pseudomonas_E sp001320045	0	100
Massilibacillus massiliensis	0	100
Macrococcus_B caseolyticus	0	100
Leuconostoc citreum	0	100
Lentilactobacillus parakefiri	0	100
Lentilactobacillus kefiri	0	100
Lactococcus raffinolactis	0	100
Lactobacillus kefiranofaciens	0	100
Lactobacillus helveticus	0	100
Klebsiella_A grimontii	0	100
Hafnia paralvei	0	100
Flavobacterium frigidarium	0	100
Enterococcus faecalis	0	100
Clostridium_B tyrobutyricum	0	100

0 25 50 75 100

Water kefir

Pseudomonas_E lactis	90.9	9.1
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Enterobacteriaceae;g__Rahnella;s__	88.9	11.1
Gluconobacter oxydans_B	86	14
Agrobacterium pusense	80	20
Gluconobacter cerinus	76.2	23.8
Pseudomonas_E peli	75	25
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Burkholderiales;f__Burkholderiaceae;g__Acidovorax_D;s__	69	31
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Burkholderiales;f__Burkholderiaceae;g__Hydrogenophaga;s__	66.7	33.3
Aeromonas rivipollensis	66.7	33.3
Aeromonas encheleia	66.7	33.3
Acinetobacter guillouiae	66.7	33.3
Acidovorax_D sp005405905	66.7	33.3
Acidovorax_D sp001633105	66.7	33.3
Acetobacter persici	58.3	41.7
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Burkholderiales;f__Burkholderiaceae;g__Aquabacterium;s__	58	42
Liquorilactobacillus ghanensis	55.6	44.4
Pantoea vagans	54.5	45.5
Gluconobacter albidus	48.9	51.1
Lacticaseibacillus paracasei	43.5	56.5
Pseudomonas_E veronii	42.9	57.1
Acetobacter indonesiensis	42.9	57.1
d__Bacteria;p__Actinobacteriota;c__Actinomycetia;o__Actinomycetales;f__Microbacteriaceae;g__Pseudoclavibacter_A;s__	42.1	57.9
d__Bacteria;p__Firmicutes;c__Bacilli;o__Lactobacillales;f__Lactobacillaceae;g__Lacticaseibacillus;s__	39.4	60.6
Acinetobacter ursingii	35.6	64.4
d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Acetobacterales;f__Acetobacteraceae;g__Komagataeibacter;s__	30.3	69.7
d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Acetobacterales;f__Acetobacteraceae;g__Gluconobacter;s__	30.1	69.9
d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Rhizobiales;f__Rhizobiaceae;g__Allorhizobium;s__	25	75
Liquorilactobacillus mali_A	12.5	87.5
Acetobacter malorum	8.7	91.3
d__Bacteria;p__Firmicutes;c__Bacilli;o__Lactobacillales;f__Lactobacillaceae;g__Liquorilactobacillus;s__	5	95
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Pseudomonadales;f__Moraxellaceae;g__Perlucidibaca;s__	0	100
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Pseudomonadales;f__Moraxellaceae;g__Alkanindiges;s__	0	100
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Aeromonadaceae;g__;s__	0	100
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Burkholderiales;f__Burkholderiaceae;g__Pelomonas;s__	0	100
d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Sphingomonadales;f__Sphingomonadaceae;g__Sphingomonas;s__	0	100
d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Caulobacterales;f__Caulobacteraceae;g__Caulobacter;s__	0	100
d__Bacteria;p__Firmicutes;c__Bacilli;o__Bacillales;f__Sporolactobacillaceae;g__Sporolactobacillus;s__	0	100
d__Bacteria;p__Actinobacteriota;c__Actinomycetia;o__Actinomycetales;f__Bifidobacteriaceae;g__Bifidobacterium;s__	0	100
Zymomonas mobilis	0	100
Sphingomonas sp002292295	0	100
Sphingomonas aurantiaca	0	100
Sphingobium sp003609795	0	100
Schleiferilactobacillus perolens	0	100
Pseudomonas_E rhodesiae	0	100
Pseudescherichia vulneris	0	100
Oenococcus oeni	0	100
Microbacterium sp000411455	0	100
Microbacterium hominis	0	100
Methylobacterium aquaticum	0	100
Liquorilactobacillus satsumensis	0	100
Liquorilactobacillus nagelii	0	100
Leuconostoc gelidum	0	100
Lentilactobacillus hilgardii	0	100
Komagataeibacter hansenii	0	100
Comamonas testosteroni_B	0	100
Comamonas acidovorans	0	100
Chryseobacterium mucoviscidosis	0	100
Caulobacter sp002742645	0	100
Brochothrix thermosphacta	0	100
Brevundimonas mediterranea	0	100
Bifidobacterium psychraerophilum	0	100
Bifidobacterium aquikefiri	0	100
Agrobacterium sp900013535	0	100
Acetobacter syzygii	0	100
Acetobacter papayae	0	100
Pseudomonas_E lundensis	25	75
Pseudomonas_E sp002966775	63.6	36.4
Leuconostoc pseudomesenteroides_A	88	12
d__Bacteria;p__Firmicutes;c__Bacilli;o__Paenibacillales;f__Paenibacillaceae;g__Paenibacillus;s__	85.7	14.3
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Enterobacteriaceae;g__Kluyvera;s__	87.5	12.5
Acinetobacter johnsonii	90.8	9.2
Lactococcus lactis_E	73.6	26.4
Raoultella terrigena	37.5	62.5
Leclercia adecarboxylata	66.7	33.3
Lactococcus lactis	87.1	12.9
Pseudomonas_E putida	61.5	38.5
Pseudomonas_E helleri	57.6	42.4
Lelliottia amnigena	60	40
Clostridium beijerinckii	91.7	8.3
Rahnella aquatilis_B	22.2	77.8
Pseudomonas_E fragi_B	21.4	78.6
Carnobacterium maltaromaticum	3.3	96.7
Acetobacter orientalis	89.3	10.7
Acetobacter fabarum	1.1	98.9
d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Pseudomonadales;f__Pseudomonadaceae;g__Pseudomonas_E;s__	0	100
d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Acetobacterales;f__Acetobacteraceae;g__Acetobacter;s__	18.8	81.2
Leuconostoc citreum	0	100
Lactobacillus kefiranofaciens	0	100
Klebsiella_A grimontii	0	100

0 25 50 75 100

Pairwise comparisons (%)